

SmartCity AR

Re-imagining Cape Town's Foreshore Freeway



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Problem Statement

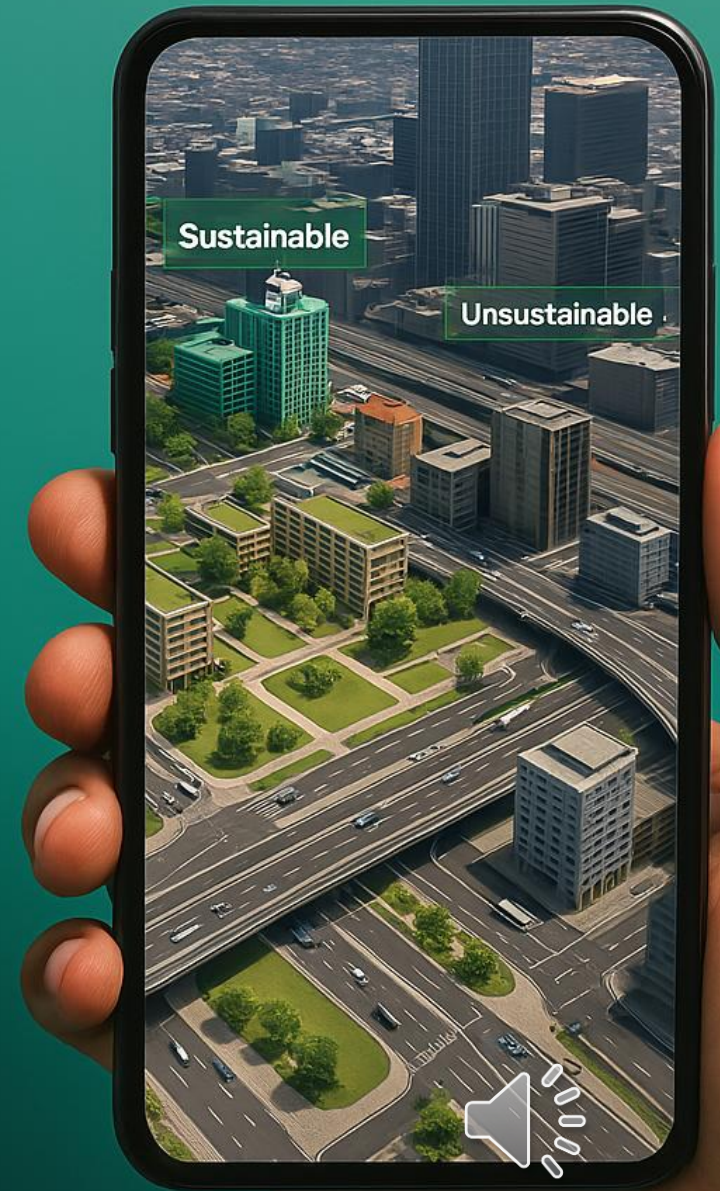
- Urban planning challenges: Traffic congestion, inefficient land use, poor pedestrian access
- Limitations of traditional planning tools: Static maps lack interactivity
- Solution: Augmented Reality (AR) for immersive city planning



WHAT IS SMARTCITY AR?

- An AR-based urban planning tool
Enhances visualization and analysis of city infrastructure
- Allows users to explore sustainable vs. unsustainable designs

Focus Area: Cape Town's
Foreshore Freeway Precinct



Key Features

- Interactive visualization of infrastructure improvements
- Toggle between planning scenarios
- Showcasing mixed-use development, MyCiTi bus routes, and other key city infrastructure
- User engagement through AR-based exploration





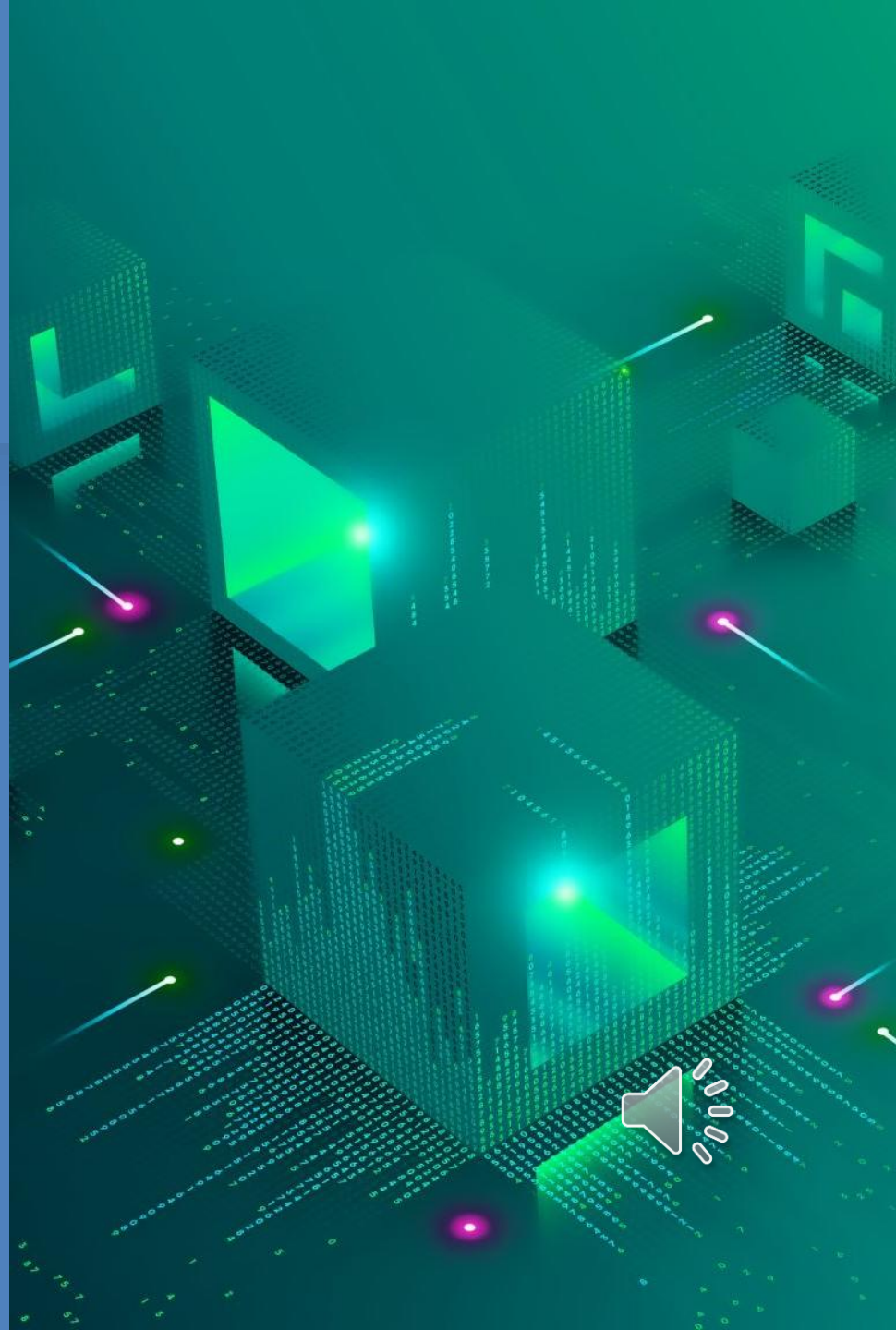
Target Users & Benefits

- Urban planners & architects: Improved project visualization
- City officials: Data-driven decision-making
- Students & researchers: Educational tool for urban studies
- Public & stakeholders: Better understanding of proposed developments



Technical Implementation

- Platform: Unity with AR Foundation (Android-based)
- Development Tools: Unity 6, Visual Studio, Blender, GitHub
- Functionality: 3D city models, interactive UI, scenario toggling



Project Scope

In Scope:

- Visualizing city layouts & planning scenarios
- Comparing sustainable vs. unsustainable development
- Basic AR interactivity & educational features

Out of Scope:

- iOS support
- Real-time data integration
- AI-based simulations
- Multiplayer/collaborative features





Conclusion

- - SmartCity AR transforms urban planning through AR
- - Provides an intuitive and interactive approach to city design

